

MLOps Assignment 3

Complete Command Reference & Project Documentation

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Project Overview

This document contains all commands and scripts used to set up and run a complete MLOps pipeline connecting Docker, Hugging Face, Kaggle, and Weights & Biases.

Component	Details
OS	Windows + WSL (Fedora Remix)
IDE	VSCode with WSL extension
Python	3.14 via virtual environment (mlops_venv)
Model	knowledgator/gliclass-llama-1.3B-v1.0
Task	Zero-shot text classification
Kaggle Competition	NLP with Disaster Tweets
Best Score	0.48605 (v3 median threshold)

1. Environment Setup

1.1 Install Docker on WSL (Fedora)

```
sudo dnf remove -y docker docker-client docker-common
sudo dnf install -y dnf-plugins-core
sudo dnf config-manager --add-repo
https://download.docker.com/linux/fedora/docker-ce.repo
sudo dnf install -y docker-ce docker-ce-cli containerd.io docker-buildx-
plugin docker-compose-plugin
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER
docker --version
docker run hello-world
```

1.2 Create Python Virtual Environment

⚠️ venv must be in WSL home (~/) not on F: drive due to NTFS permission issues

```
python3 -m venv ~/mlops_venv
source ~/mlops_venv/bin/activate
pip install ipykernel jupyter numpy pandas matplotlib huggingface_hub
transformers kaggle wandb torch gliclass flask
python -m ipykernel install --user --name=mlops_venv --display-name
"Python (MLOps-A3)"
```

1.3 Open VSCode in WSL Mode

```
cd "/mnt/f/MTechAI_ML_IITJ/Semester2/ML
Ops/Assignments/Assignment_3/work/MLOps_Assignment3V1"
code .
```

⚠ *Always open VSCode this way — from WSL terminal using 'code .' to keep WSL mode active*

1.4 Daily Workflow (Run Every Session)

```
source ~/mlops_venv/bin/activate
cd "/mnt/f/MTechAI_ML_IITJ/Semester2/ML
Ops/Assignments/Assignment_3/work/MLOps_Assignment3V1"
code .
```

2. Hugging Face Setup

2.1 Install & Login

```
pip install huggingface_hub
hf auth login
# Paste your token when prompted (starts with hf_)
hf auth whoami
```

2.2 Download Model

⚠ *Model is saved to F: drive to save WSL home space*

```
from huggingface_hub import snapshot_download, login

login(token="hf_your_token_here")

model_path = snapshot_download(
    repo_id="knowledgator/gliclass-llama-1.3B-v1.0",
    local_dir="/mnt/f/models/gliclass-llama-1.3B"
)
print("Downloaded to:", model_path)
```

2.3 Check Model Files

```
ls /mnt/f/models/gliclass-llama-1.3B
```

3. Kaggle Setup

3.1 Create kaggle.json Manually

⚠️ Go to <https://kaggle.com/settings> -> API -> Create New Token to get your key

```
mkdir -p ~/.kaggle
cat > ~/.kaggle/kaggle.json << 'EOF'
{"username":"ausafiitj","key":"YOUR_KAGGLE_KEY_HERE"}
EOF
chmod 600 ~/.kaggle/kaggle.json
cat ~/.kaggle/kaggle.json
```

3.2 Test Kaggle Connection

```
kaggle datasets list
```

3.3 Download Competition Data

```
kaggle competitions download -c nlp-getting-started -p
/mnt/f/datasets/disaster-tweets
unzip /mnt/f/datasets/disaster-tweets/nlp-getting-started.zip -d
/mnt/f/datasets/disaster-tweets/
ls /mnt/f/datasets/disaster-tweets/
```

4. Weights & Biases Setup

4.1 Login to W&B

⚠️ Get API key from <https://wandb.ai/authorize> — must be 40+ characters

```
pip install wandb
wandb login --relogin
# Paste your API key when prompted
```

4.2 Test W&B Connection

```
python3 /tmp/test_wandb.py
```

Create /tmp/test_wandb.py with:

```
import wandb
```

```
wandb.login()
run = wandb.init(project='mlops-assignment3', mode='online')
print('Run ID:', run.id)
run.finish()
```

5. Docker Configuration

5.1 Enable Docker in WSL

- Open Docker Desktop on Windows
- Go to Settings -> Resources -> WSL Integration
- Enable 'Enable integration with my default WSL distro'
- Toggle ON FedoraRemix
- Click Apply & Restart

```
docker --version
docker run hello-world
```

5.2 Dockerfile

```
FROM python:3.11-slim

WORKDIR /app

RUN pip install --no-cache-dir \
    gliclass \
    torch \
    huggingface_hub \
    wandb \
    pandas \
    flask

COPY run_model.py .

CMD ["python", "run_model.py"]
```

5.3 Build Docker Image

```
cd "/mnt/f/MTechAI_ML_IITJ/Semester2/ML
Ops/Assignments/Assignment_3/work/MLOps_Assignment3V1"
docker build -t gliclass-model .
docker build --no-cache -t gliclass-model . # force fresh build
```

6. Model Scripts (run_model.py Evolution)

6.1 Version 1 — Basic Test

First working test with single text classification:

```
from gliclass import GLiClassModel, ZeroShotClassificationPipeline
from transformers import AutoTokenizer
import torch

model_path = "/models/gliclass-llama-1.3B"
tokenizer = AutoTokenizer.from_pretrained(model_path)
model = GLiClassModel.from_pretrained(model_path,
torch_dtype=torch.float32)
model = model.float() # Fix BFloat16 vs Float32 dtype mismatch error

pipeline = ZeroShotClassificationPipeline(
    model, tokenizer,
    classification_type='multi-label',
    device='cpu'
)

text = "Apple announces new iPhone with amazing features"
labels = ["technology", "sports", "politics", "business"]
result = pipeline(text, labels, threshold=0.1)
print("Result:", result)
```

6.2 Version 2 — With W&B Logging

```
import wandb
import torch
from gliclass import GLiClassModel, ZeroShotClassificationPipeline
from transformers import AutoTokenizer

run = wandb.init(project='mlops-assignment3', name='gliclass-inference',
    config={"model": "gliclass-llama-1.3B", "threshold": 0.1, "device":
"cpu"})

model_path = "/models/gliclass-llama-1.3B"
tokenizer = AutoTokenizer.from_pretrained(model_path)
model = GLiClassModel.from_pretrained(model_path,
torch_dtype=torch.float32)
model = model.float()

pipeline = ZeroShotClassificationPipeline(model, tokenizer,
    classification_type='multi-label', device='cpu')
```

```

text = "Apple announces new iPhone with amazing features"
labels = ["technology", "sports", "politics", "business"]
result = pipeline(text, labels, threshold=0.1)

scores = {}
for item in result:
    for pred in item:
        print(f" {pred['label']}: {pred['score']:.4f}")
        scores[pred['label']] = pred['score']

wandb.log(scores)
run.finish()

```

6.3 Version 3 — Kaggle Disaster Tweets (v1, threshold=0.3)

⚠ *Problem: threshold=0.3 too low — ALL 3263 tweets classified as disaster (score=1.0)*

```

# threshold=0.3 caused every tweet to be predicted as disaster
prediction = 1 if disaster_score > 0.3 else 0    # BAD - too low
# Kaggle score: 0.42966

```

6.4 Version 4 — Compare Labels Directly (threshold fix attempt)

⚠ *Problem: comparing disaster vs not_disaster scores — ALL tweets got 0 (not disaster)*

```

# Compare scores directly - still wrong
prediction = 1 if disaster_score > not_disaster_score else 0    # BAD
# Result: total_disaster=0, total_not_disaster=3263

```

6.5 Version 5 — FINAL: Median Threshold (Best Score 0.48605)

⚠ *Solution: Use median score as dynamic threshold for balanced predictions*

```

import torch
import wandb
import pandas as pd
import statistics
from gliclass import GLiClassModel, ZeroShotClassificationPipeline
from transformers import AutoTokenizer

run = wandb.init(project='mlops-assignment3', name='kaggle-disaster-
tweets-v3')

tokenizer = AutoTokenizer.from_pretrained("/models/gliclass-llama-1.3B")
model = GLiClassModel.from_pretrained("/models/gliclass-llama-1.3B",

```

```

    torch_dtype=torch.float32)
model = model.float()

pipeline = ZeroShotClassificationPipeline(model, tokenizer,
    classification_type='multi-label', device='cpu')

test_df = pd.read_csv("/data/test.csv")
labels = ["disaster", "not disaster"]
disaster_scores = []

for i, row in test_df.iterrows():
    text = str(row['text'])
    result = pipeline(text, labels, threshold=0.01) # very low to get all
    scores = {pred['label']: pred['score'] for item in result for pred in
    item}
    disaster_score = scores.get('disaster', 0)
    disaster_scores.append(disaster_score)
    if i % 100 == 0:
        wandb.log({"progress": i, "disaster_score": disaster_score})

# Use median as dynamic threshold – gives balanced 50/50 predictions
median_score = statistics.median(disaster_scores)
print(f"Median disaster score: {median_score:.4f}")

predictions = [1 if s > median_score else 0 for s in disaster_scores]

submission = pd.DataFrame({'id': test_df['id'], 'target': predictions})
submission.to_csv("/data/submission.csv", index=False)
print(submission['target'].value_counts())

wandb.log({"median_threshold": median_score,
    "total_disaster": sum(predictions),
    "total_not_disaster": len(predictions) - sum(predictions)})
run.finish()

```

7. Docker Run Commands

7.1 Basic Model Run

```

docker run --rm \
    -v "/mnt/f/models/gliclass-llama-1.3B:/models/gliclass-llama-1.3B:ro" \
    gliclass-model

```

7.2 With W&B Logging

```
docker run --rm \
-v "/mnt/f/models/gliclass-llama-1.3B:/models/gliclass-llama-1.3B:ro" \
-v ~/.netrc:/root/.netrc:ro \
gliclass-model
```

7.3 With Kaggle Data + W&B

```
docker run --rm \
-v "/mnt/f/models/gliclass-llama-1.3B:/models/gliclass-llama-1.3B:ro" \
-v "/mnt/f/datasets/disaster-tweets:/data" \
-v ~/.netrc:/root/.netrc:ro \
gliclass-model
```

7.4 Web Interface (Flask)

```
docker run --rm \
-p 0.0.0.0:8080:8080 \
-v "/mnt/f/models/gliclass-llama-1.3B:/models/gliclass-llama-1.3B:ro" \
-v ~/.netrc:/root/.netrc:ro \
gliclass-model
# Then open: http://localhost:8080
```

8. Kaggle Submission

8.1 Submit to Competition

```
source ~/mlops_venv/bin/activate
kaggle competitions submit -c nlp-getting-started \
-f /mnt/f/datasets/disaster-tweets/submission.csv \
-m "GLiClass zero-shot v3 median threshold"
```

8.2 Check Submission Score

```
kaggle competitions submissions -c nlp-getting-started
```

8.3 Scores Achieved

Version	Approach	Score	Notes
v1	Fixed threshold=0.3	0.42966	All 3263 tweets = disaster
v3 	Median threshold	0.48605	Balanced: 1631 disaster, 1632 not

9. Useful Diagnostic Commands

9.1 Check Everything is Working

```
# Check Python
python3 --version
which python3

# Check venv
source ~/mlops_venv/bin/activate
pip list | grep -E 'torch|transformers|gliclass|wandb|kaggle'

# Check Docker
docker --version
docker ps

# Check Kaggle
kaggle datasets list

# Check W&B
cat ~/.netrc

# Check HF
hf auth whoami
```

9.2 Fix Common Issues

```
# Fix: kaggle command not found
source ~/mlops_venv/bin/activate

# Fix: docker permission denied
newgrp docker

# Fix: WSL drive not mounted
sudo mkdir /mnt/f
sudo mount -t drvfs F: /mnt/f

# Fix: VSCode not in WSL mode
# Open WSL terminal -> cd to project -> code .

# Fix: HF token invalid
import os
os.environ.pop('HF_TOKEN', None) # remove old bad token
from huggingface_hub import login
```

```
login(token='hf_your_token')
```

10. W&B Dashboard

View all experiment runs and metrics at:

```
https://wandb.ai/ausaf-ahmad-iitj3-indian-institute-of-technology-jodhpur/mlops-assignment3
```

Runs logged:

- curious-deluge-2 — first W&B test run
- gliclass-inference — basic classification with scores
- kaggle-disaster-tweets — v1 all disaster
- kaggle-disaster-tweets-v2 — v2 all not disaster
- kaggle-disaster-tweets-v3 — v3 median threshold (BEST)

11. Final Setup Summary

Service	Status	Details
WSL + Fedora Remix	✓ Done	Fedora Remix on Windows WSL
VSCode + Jupyter	✓ Done	WSL mode, Python (MLOps-A3) kernel
Virtual Environment	✓ Done	~/mlops_venv
Hugging Face	✓ Done	Model downloaded to /mnt/f/models/
Docker	✓ Done	WSL integration enabled
Kaggle	✓ Done	Authenticated as ausafitj
W&B	✓ Done	Project mlops-assignment3
Model in Docker	✓ Done	GLiClass running in container
Kaggle Submission	✓ Done	Best score: 0.48605